Term Lengths and Parliamentary Performance: Evidence from Natural Experiments in India

Abstract

Does the quality of representatives shape how they respond to institutional incentives? The duration of a political term is a powerful incentive that can change representatives' priorities and shape their performance. We argue that positive selection of representatives can override the negative incentives created by shorter terms. We demonstrate this by leveraging natural experiments in the indirectly-elected *Rajya Sabha* of the Indian Parliament. In 1952, a public lottery randomly assigned term lengths to a founding cohort of high-quality representatives. We combine this setup with detailed data from their parliamentary records to show that shorter terms did not worsen parliamentary performance of these representatives. Second, we leverage quasi-random variation in term lengths following the unexpected deaths of sitting representatives. We find that a shorter term significantly worsens parliamentary performance once positive selection is much harder. Taken together, these findings indicate that the effects of perverse incentives are conditional, and depend on the type of actors they operate upon.

Word count: 14184

1 Introduction

Term lengths—the length of period that representatives serve before contesting for re-election—play an important role in shaping incentives in legislatures around the world. There is a large variation in the length of terms that elected representatives enjoy—both across[1] and within countries[2]. Following the expiration of a term, all members are subject to elections, with no guarantee of returning to office. If terms are too short, incumbents may prioritize election-related activities at the cost of policy-making. On the other hand, longer terms can diminish electoral accountability and responsiveness through infrequent elections.

There are theoretical reasons to believe that term lengths can change representatives' behaviors and priorities (Schultz 2008; Gersbach et al. 2020). A limited time frame can compel politicians to reallocate their efforts to focus on achieving short-term political goals. Yet, such changes in incentives may not meaningfully alter the behavior of self-motivated and positively selected representatives: different *types* of politicians can respond to incentives in different ways (Callen et al. 2025; Galasso and Nannicini 2011).

Empirically testing such theories present several limitations. First, opportunities to study the effect of term lengths are scarce. We only observe how legislators behave once they have been assigned a particular term length and these are fixed for different legislatures. Given these constraints, researchers might turn to cross-national analyses, however, these suffer from unmeasured confounding.

Second, the literature on term lengths has shown mixed results: shorter terms worsen performance and outcomes in some cases but not others (Titiunik 2016; Dal Bo and Rossi 2011; Fukumoto and Matsuo 2015; Gaines et al. 2012; Pomirchy 2023). This suggests the perverse effects of electoral incentives could be conditional, moderated by the quality of representatives. Consequently, disentangling these dynamics requires empirical settings that allow researchers to evaluate selection and incentives simultaneously.

This paper addresses both limitations. First, we introduce a theory of how the incentive of term

[1] For a non-exhaustive summary see Supplementary Index of Titiunik (2016).

[2] For instance, in the United States, members of the House of Representatives enjoy a two-year term, whereas members of the Senate enjoy six-year terms.

lengths interacts with selection. We focus on one avenue of representatives' behavior that is public and consequential for accountability—that is, on parliamentary performance. In line with the selection model of political representation (Mansbridge 2009), we expect that the positive selection of representatives overcomes the perverse incentives of shorter terms, and thus does not undermine their parliamentary performance.

We test our theoretical propositions by leveraging a natural experiment in the indirectly elected *Rajya Sabha*, a newly-formed legislature following India's independence movement. This setting provides an ideal opportunity to study the effect of term lengths on a high quality of representatives. The newly formed institution was predominantly composed of experienced, highly educated politicians, many of whom had previously served in provincial parliaments or on the constitutional drafting committee. Moreover, their central involvement in the independence movement suggests that their political careers were shaped by strong ideological and intrinsic motivations. In November of 1952, a public lottery was used to randomly assign the members of the newly composed *Rajya Sabha* to two-, four-, and six-year terms. This was done to stagger the entry and exit of members in a house where one-third of its members were to retire every six years. To the best of our knowledge, this is the first time this causal setting has been utilized, and given its clean design, can be leveraged to study how term lengths affected parliamentary performance.[3] Archival reports and balance tests rule out any manipulation in the experiment. Importantly, elected members in the Indian parliament are not subject to term limits, which allows us to attribute variation in performance to term lengths alone.

We test the hypothesis that shorter terms affects parliamentary performance by collecting daily data on members' participation in parliamentary activities. We draw on a novel dataset of almost 400,000 verbatim debates and discussions conducted in the *Rajya Sabha* to quantify individual effort for all elected members over a forty year period. Particularly, we measure each representatives' participation in the question hour, submission of written questions, and participating in discussions on matters of public importance.

The empirical results support our expectation: the high quality of the founding members overcomes

[3] Such a staggered entry and exit of members is also observed in the French Senate, Kazakhstan's Senate, Pakistan's Senate and Nepal's National Assembly, and can be similarly leveraged to study the effects of shorter terms.

the negative incentives created by shorter terms. We do not find evidence that a shorter term worsens the parliamentary performance of members. Point estimates remain small and statistically insignificant. We conduct power calculations to show that the experiment has sufficient power to detect small effect sizes, and equivalence tests rule out effect sizes larger than 0.2 standard deviations. Further, flexible explorations of heterogeneity finds no meaningful sub-groups and effects do not vary over time, by election proximity. Taken together, these results show that shorter terms do not affect the performance of these high quality representatives.

An alternative interpretation of this null finding could be that the indirect mode of election might be sufficiently insulating representatives from electoral pressure. In this scenario, term lengths would fail to predict performance for *any* member, irrespective of selection. To isolate the role of selection in the same house, we leverage a second research design, focusing on a cohort where ensuring positive selection is much harder. We take advantage of a quasi-random variation in term lengths arising due to unexpected deaths of sitting members of parliament (MPs). In this set up, we find that shorter terms significantly worsen the performance of the "replacement" MPs, reducing the average daily participation by 64-75% across all measures. These results remain robust to sources of unmeasured confounding and are not driven by our estimation choices.

Taken together, the results highlight how selection interacts with perverse incentives to shape members' performance. This paper makes important contributions to the existing literature, by building on research that focuses on the determinants of the performance of elected representatives (Fong 2020; Kellermann 2009; Grumbach and Sahn 2020; Fagan and McGee 2020; Napolio and Grose 2021; Zelizer 2019; Ferraz and Finan 2009), by providing evidence on the *conditions* under which term lengths can shape outcomes (Titiunik 2016; Dal Bo and Rossi 2011).

Relatedly, this paper contributes to a rich set of studies that concern dynamics in legislatures whose members are indirectly elected. Over a tenth of legislatures across the world are indirectly elected by sub-national legislatures or local councils. The indirect mode of election has shown to decrease electoral accountability and performance (Micozzi 2013), lower responsiveness towards the electorate (Gailmard and Jenkins 2009) and lead to a moderation of ideology (Bernhard and Sala 2006). Our paper contributes to these studies by highlighting how institutional incentives can operate in an indirectly elected legislature to shape performance. Specifically, we bring together the literature on political representation (Mansbridge 2009), indirectly-elected legislatures (Bhatia 2022) and quality of representatives (Gulzar 2021; Gulzar and Khan 2025; Chaudhuri et al. 2025; Galasso and Nannicini 2011) to argue that clarifying the dominant model of political representation remains central to understanding how incentives shape performance.

Empirically, we contribute to the literature on experiments and research designs within political institutions that attempt to study the behavior of political elites (Grose 2021). Specifically, our study introduces new identification strategies that can be employed to analyze the effects of term lengths, that are adaptable across a range of comparative contexts. In doing so, we are able to offer a framework for examining how a particular feature of institutional design—the length of term in office—shapes the incentives and behaviors of elected representatives.

2 Theoretical Framework

2.1 Selection and Incentives

Two models of political representation have been pointed out in the political science literature (Mansbridge 2009). The first is a model based on *sanctions*, wherein the principals (voters) need to monitor their agent (the elected representative) to reward or punish them. The second is a model based on *selection*, wherein the principles and agents are aligned in their objective, even in the absence of incentives and sanctions, as the agent is internally motivated to pursue common goals. Clarifying the dominant model of political representation at play is the key to understanding how incentives affect the behaviors. When representation aligns with the selection model, agents' intrinsic motivations can be sufficient to overcome any negative incentives in the institutional set-up.

Early work modeling the effects of term lengths on parliamentary and legislative behavior propose that shorter terms can make politicians distort their policy preferences (Schultz 2008) and a small number of papers have tested the general negative effects of shorter terms that involve shirking on account of campaigning activities (Titiunik 2016; Gaines et al. 2012; Dal Bo and Rossi 2011), resulting in changes in policy support (Gersbach et al. 2020) and responsiveness (Titiunik 2016; Pomirchy 2023).

More specifically, the negative incentives of shorter terms on parliamentary activities arise through three mechanisms, which do not apply to high quality members. The first mechanism is that shorter terms reduce performance due to shirking from legislative activities, as members divert time to conduct campaigns (Titiunik 2016; Dal Bo and Rossi 2011). High quality members may already possess political capital, visibility and recognition, that reduces campaign costs. A second mechanism is that shorter terms reduces the policy horizon which changes the *type* of activities taken up, however, high quality members could have sufficient intrinsic motivation to take on complicated policy issues inspite of the reduced horizon. The third mechanism is based on a lack of capacity: while first-time members face a steep learning curve to learn about parliamentary procedure, high quality members could already have this experience, and a lack of capacity does not hamper their performance.

Even when members serve *constant* terms, temporal proximity can change behaviors: members may reduce their participation in parliamentary activities as elections approach, a pattern that may not hold if elected representatives are of a higher quality. Together, high quality members serving shorter terms need not change their behaviors, in response to incentives or over time, to accommodate their political goals.

2.2 Indirectly Elected Legislatures

These patterns are especially applicable to the case of *indirectly elected legislatures*. Over a tenth of legislatures across the world are indirectly elected by sub-national legislatures or local councils: including the *German Bundesrat*, whose members are delegated by the respective state governments, and the *French Senate*, whose members are elected indirectly by 150,000 officials. We describe the phenomenon of indirectly elected houses using data on national legislatures from the Inter-Parliamentary Union (IPU). While directly-elected houses are more common across countries, we observe the prevalence of indirectly-elected houses, predominantly as a part of a bicameral legislature, across countries across the Global North and South, as seen in Table A1.

These legislatures are typically elected by subnational units like provincial assemblies or municipal councils, which form the selectorate. At times, a fixed proportion of members are appointed directly by the head of the state. Countries continue to experiment with directly and indirectly elected houses—for instance, Belgium passed legislation in 2014 to make its Senate composed of indirectly elected members, and policies that elect the head of village councils in Maharashtra (India) based on direct or indirect elections have seen three reversals in the last decade (Heinze 2025). There is considerable variation in the length of terms across these indirectly-elected houses. In a majority of these legislatures, members are elected for 6-year term (60%), while in other legislatures, members are elected for 5-year terms (40%) or lower.

Indirect elections shape representation and the chain of delegation in important ways. When representatives are elected through an indirect election, the mass electorate (principal) delegates responsibility to the representative (agent) in two steps. The principals first select an agent (the intermediary electors), who further selects another agent, the national-level representative. Intermediary electors are typically members who hold political office at a lower level, and could be better at selecting and monitoring national representatives than the mass electorate (Gailmard and Jenkins 2009). Further, if intermediary electors can be strongly influenced by party leaders, national representatives would be directly accountable and responsive to the concerns of the party leadership rather than the mass electorate (Rogers 2012).

2.3 Selection Model in an Indirectly Elected Legislature

Historically, indirect elections have been justified using two rationales, that also follow from the selection model: first, indirectly elected members are characterized by superior quality that enhances the competence of the legislature and second, they may be desirable in young democracies with lower levels of development wherein voters are illiterate and may not be able to elect the "right kinds of men" (Bhatia 2022). Indirectly elected members go through "successive filters" (Bhatia 2022), wherein first citizens elect members to sub-national legislatures, and these further elect members to the national parliament. This process results in the election of candidates with greater experience and strong intrinsic motivations.

In addition to experience and competence, indirectly elected representatives are chosen through a process that emphasizes internal party support. As an intermediary body of elected members chooses representatives in an indirectly elected legislature, such legislators may have already had longer and more influential political careers. These dynamics can also ensure that such members are sufficiently insulated from public pressures, which can help them prioritize long-term policy goals.

For these reasons, negative incentives produced by shorter terms may not affect the actions of elected members in an indirectly elected house generally. The indirect mode of election may provide sufficient guardrails, and members elected to an indirectly elected house will continue to prioritize parliamentary performance which furthers their political goals, without being affected by perverse incentives created by shorter terms.

3 Institutional Background and Data

In this section, we describe the institutional details of the indirectly-elected house of the Indian Parliament. The *Rajya Sabha* is a federal chamber established in 1952[4], that is comprised of members who are indirectly elected by elected representatives of sub-national (state-level) legislatures. It sits alongside a directly elected federal chamber called the *Lok Sabha*, which enjoys some greater powers in law-making over the *Rajya Sabha*, particularly pertaining to financial matters[5]. Beyond fiscal legislation, members of the two Chambers enjoy similar powers, privileges and responsibilities.

The strength of the *Rajya Sabha* has gradually increased over the years—from 216 members in 1952 to 245 members in 2024. A small share of members are not elected but nominated by the President of India. The entire house is not subject to dissolution at the same time—members serve for six-year terms, with one third of the members vacating their positions every second year. As vacancies arise, state-level assemblies associated with the vacant seats elect incumbent or new representatives to fill the seats. Members of the *Rajya Sabha* are not subject to any term limits, allowing for multiple successive terms.

Members of the *Rajya Sabha* have a number of parliamentary tools available to them to facilitate their participation in the daily proceedings of the House. We describe these tools below and how

[4] In 1952, the first Parliament of "Independent India" came into existence, two years after the Indian Constitution was officially adopted with which India became a Republic after over 200 years of British Colonial Rule. A unicameral Constituent Assembly first met in 1946 and served as a legislature till the first elections were held in 1952. The utility of a second chamber was debated, and ultimately, the assembly decided to have a bicameral legislature. Thus the *Rajya Sabha* was created, whose mode of election and composition differed from the directly elected 'House of People' (Election Commission of India 1955).

[5] In particular, only the *Lok Sabha* can pass or modify money bills that regulate taxes and government spending such as the annual budget. The Council of Ministers is only responsible to the *Lok Sabha* and these members alone can initiate no-confidence and adjournment motions among others to discipline the executive branch.

we use them to measure legislative performance. We focus on parliamentary devices that rely on individual effort with the goal of exercising oversight on the functioning of the executive and the policies of the government.

Parliamentary Questions: Parliamentary questions allow elected members to exact responses regarding government policies in the public record, as well as elicit information on the working of the administration and implementation of policies. Government ministers are supposed to provide authentic and correct answers to these questions. These questions are either answered orally during the first hour of each house meeting (during the Question Hour) or through written answers. If the number of submitted questions exceeds the daily quota, lots are drawn to select the questions that will be answered. We quantify members' efforts by measuring two activities: first, the number of times members engage in discussions during the question hour, and second, the number of written questions raised by them.

Participation in Discussions: Members can use parliamentary devices to initiate and participate in discussions in the house. This includes short duration discussions and half-hour discussions that draw attention of the Government to matters of urgent or sufficient public importance, and motions on matters of public interest. We measure the number of times each member participates in these discussions.

Most bills are written and introduced by the ruling party, and elected members contribute to law-making by participating in debates and discussions on bills. Despite the substantive importance of such debates in the legislative process, we exclude them from our measures of parliamentary performance. Floor participation during debates is strictly controlled by party leaders and party whips: and consequently, they are not reflective of individual effort.

Taken together, we measure parliamentary performance by looking at how often members exercise oversight through questions, and draw the attention of the government to matters of public importance through discussions. The primary source for these activities is the online corpus of *Official Debates of the Rajya Sabha*: an institutional platform operated by the *Rajya Sabha Secretariat*, the non-partisan executive wing of the chamber. Daily and historical proceedings can be accessed through website, which makes both the metadata and text of the debates publicly available. We scrape the metadata and raw text of over 400,000 debates between 1952 and 1995. Each record captures date, subject, a short title and participating members. This granular daily data allows us to analyze participation both in aggregate and over time.

We merge these granular data on performance with detailed member characteristics digitized from the *Rajya Sabha Who's Who* published by the *Rajya Sabha Secretariat* in 1952 and 2019. These biographies allow us to capture a rich set of covariates for each elected member: including age, gender, education, professional experience, party affliation and political positions held by them.

4 Research Design

As introduced in the previous section, members of the Rajya Sabha are elected to six-year terms. To maintain the chamber as a continuous body, one-third of its members must retire every two years. Implementing this staggered rotation in the inaugural Parliament posed a unique problem, which was resolved by drawing lots in public. For our main research design, we leverage this public lottery a natural experiment.[6]

Once elected, the first Rajya Sabha met in April 1952 to conduct its activities.[7] The Council of States (Term of Office of Members) Order stipulated when and how lots were to be drawn in public to determine the term lengths for each member. Importantly, the rules for term length refer only to the expiration of the term that require members to vacate their seat. These members were not term-limited and could freely seek re-election.

The order stipulated how many members from each state would be placed across the three groups of members, who were to vacate their seats after two, four or six years, as depicted in Table 1. Of the 216 members in the Rajya Sabha, the members representing the regions of *Ajmer and Coorg* and *Manipur and Tripura* were assigned a fixed two-year term, while 72 members were to be assigned a six-year term, and 71 members assigned to a four-year or a two-year term. Approximately six

[6]Related literature on term lengths has looked at other sources of randomization in legislative chambers as a result of redistricting (Titiunik 2016) or constitutional reforms (Dal Bo and Rossi 2011).

[7]The first elections were held in 1951 and early 1952 for the Lok Sabha and various state assemblies. For Jammu and Kashmir, representatives were chosen by the President on recommendation of the State Government. For Kutch and Tripura, members were elected through electoral colleges. On the 4th March, 1952, these elected members from state-level bodies elected the Rajya Sabha. These elections were completed by the end of March, 1952. The names of all winners were published in the Gazette of India on the 31st of March, 1952, while the names of the nominated members were published on 2 April 1952 (Election Commission of India 1955).

Table 1: Lottery determines number of members in each group across states

State	6-Year Term	4-Year Term	2-Year Term
Assam	2	2	2
Bihar	7	7	7
Bombay	6	6	5
Madhya Pradesh	4	4	4
Madras	9	9	9
Orissa	3	3	3
Punjab	3	2	3
Uttar Pradesh	10	11	10
West Bengal	5	4	5
Hyderabad	3	4	4
Jammu and Kashmir	2	1	1
Madhya Bharat	2	2	2
Mysore	2	2	2
P.E.P.S.U.	1	1	1
Rajasthan	3	3	3
Saurashtra	1	2	1
Travancore-Cochin	2	2	2
Vindhya Pradesh	1	1	2
Bhopal, Bilaspur, Himachal Pradesh, Delhi, and Kutch	2	1	1
Nominated by the President	4	4	4
Total	**72**	**71**	**71**

Note: Digitized version of Schedule 3 in the Council of States (Term of Office of Members) Order, 1952 that displays the number of members by states to be assigned to six, four or two-year terms.

months after taking office, the public lottery was conducted and the names of members assigned to the different groups were published in the Official Gazettes in November 1952.

To validate our identification strategy, we test for potential manipulation in the lottery process—a salient concern given the high stakes for members of the first Parliament. Specifically, longer terms might have been disproportionately assigned to influential members of the ruling party, or high performers during the first two legislative sessions before the lottery was conducted. To rule out such selection bias, we examine newspaper archives of parliamentary reporting from 1952 to 1953 and conduct comprehensive balance tests. Our results demonstrate that the term lengths were strictly randomized without any manipulation.

No evidence of manipulation in historical records: We searched the archives the Times of India (1952–1953) for coverage of the public lottery that assigned members different term lengths. While we found articles discussing the public ballot, none reported any manipulation. Furthermore, the paper's "Week in Parliament" bulletin[8] recorded no protests or controversies regarding the process. We even found reports of a Chief Minister receiving a short term in a similar state-level lottery,[9] demonstrating that senior members were not exempt from unfavorable draws. This absence of public contestation strengthens our confidence in the integrity of the random assignment.

Evidence from balance tests: Balance in the background characteristics of MPs and pre-treatment measures of our outcomes provide further credibility to the research design. We code background characteristics from the member biographies and construct pre-treatment measures of parliamentary activities during the two sessions before the lottery assigned term lengths to each member. This includes their participation in the question hour, written questions submitted, discussions participated in and an index of the three measures.[10] The data allow us to measure competence, a dimension of political quality (Chaudhuri et al. 2025; Galasso and Nannicini 2011), and observe that overall, the members are highly educated and served in important political and leadership positions as a part of the independence movement.

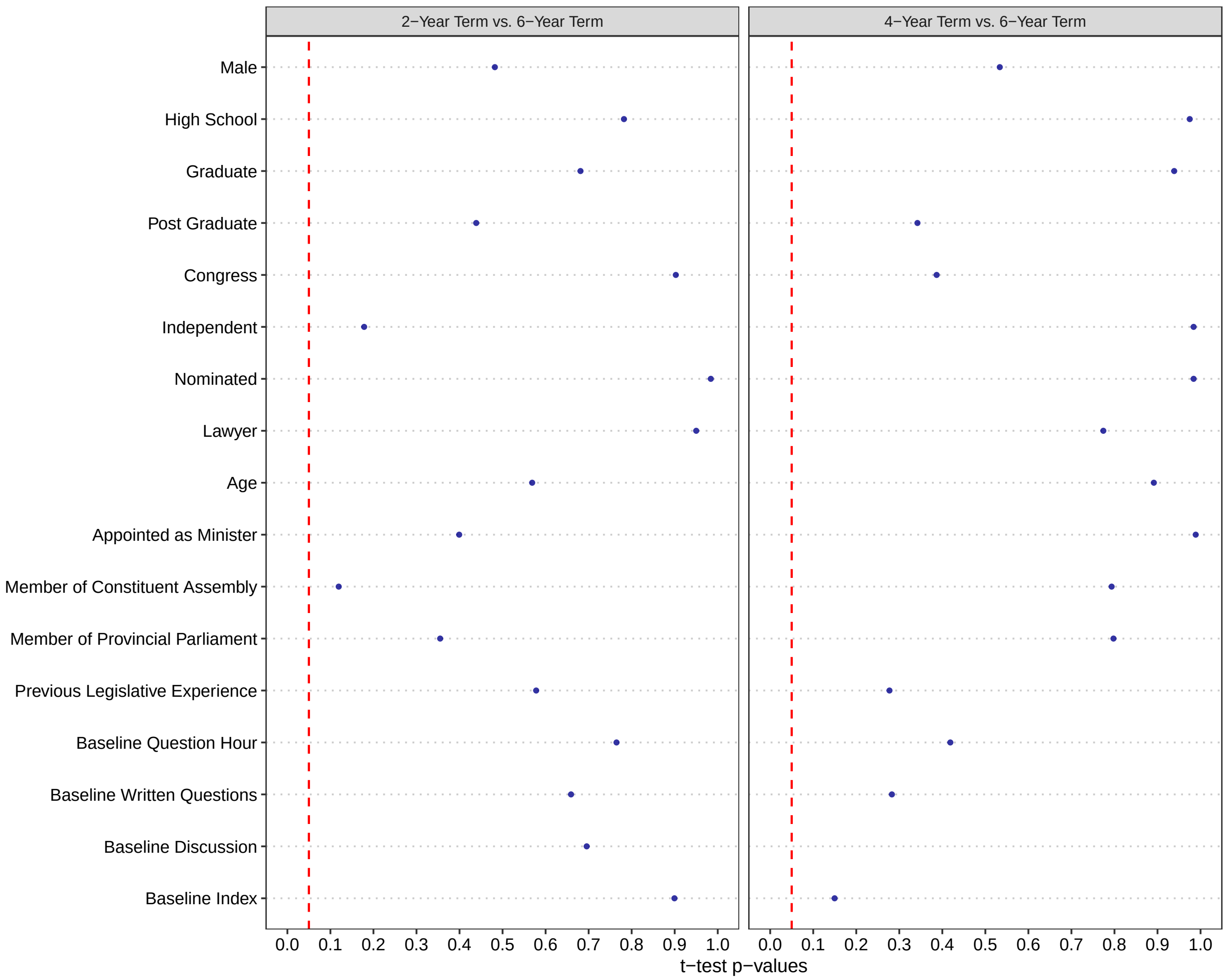
We find that the three groups are balanced, as seen in Figure 1. We see that the coefficients are balanced across all covariates, and p-values associated with the t-tests are not statistically significant. An F-test shows that these predictors do not jointly explain the treatment: we regress the predictors on a binary version of the treatment of shorter terms (2-Year or 4-Year term), the 2-Year treatment and the 4-Year treatments individually in a restricted model, and compare it to an unrestricted model. The p-values associated with the Partial F-Test are 0.82 (binary treatment), 0.58 (2-Year treatment) and 0.91 (4-Year treatment). We conclude that the predictors do not jointly explain any of the treatments, which provides further credibility to the clean randomization from the lottery.

[8] Week in Parliament: The Council Protests, Parliament House, BG, The Times of India (1861-2010); Nov 30, 1952; ProQuest Historical Newspapers: The Times of India pg. 11.

[9] Membership of the State Council: 4 Years For "C.R." The Times of India (1861-2010); Nov 20, 1952; ProQuest Historical Newspapers: The Times of India page 8.

[10] In particular, only the *Lok Sabha* can pass or modify money bills that regulate taxes and government spending such as the annual budget. The Council of Ministers is only responsible to the *Lok Sabha* and these members alone can initiate no-confidence and adjournment motions among others to discipline the executive branch.

Figure 1: Lottery balances MPs across treatment and control groups



Note: Dotted red line indicates the 0.05 threshold for the p-values from t-tests. Variables are indicators, except *Age* and *Previous Legislative Experience* and the five pre-treatment measures of legislative activities which are measured in number of years. Table A2 displays exact group means and p-values.

5 Empirical Results

We use the lottery to test if shorter term lengths worsen parliamentary performance. 72 members in the control group served the stipulated term of six years, whereas other members were assigned shorter terms: 71 members were to serve either a two-year or four-year term. The data confirms that the treatment worked as intended. Almost all MPs vacated their offices in line with the rule, except those who passed away, vacated their seats to hold another elected position or resigned. Because actual time served was subject to these natural interruptions, we interpret our results as Intent-to-Treat (ITT) estimates based on the initial lottery assignment.

To estimate the average treatment effects of shorter term lengths on parliamentary performance, we employ a linear regression model. We improve the precision of our estimates by controlling for baseline (pre-treatment) performance measures during the six-month period prior to the lottery. We analyze this as a block randomized experiment with two treatment arms, evaluating outcomes aggregated over the first two years of the Rajya Sabha when all three groups were present in the chamber. We estimate treatment effects with the following specification:

$$Y_{ij} = \beta_0 + \beta_1 T1_{ij} + \beta_2 T2_{ij} + \gamma X_{ij} + \delta_j + \epsilon_{ij} \tag{1}$$

Y_{ij} denotes outcomes of interest for MP i from state j over the two year period, capturing their total participation in the question hour, written questions submitted and discussions participated. We also create an equally weighted index of the three outcomes. The variables $T1_{ij}$ and $T2_{ij}$ are binary indicators for assignment to the two-year and four-year treatment arms, respectively. To account for experimental blocking, δ_j represents a state fixed effects. X_{ij} is a pre-treatment measure of the outcome and ϵ_{ij} is the error term. The coefficients β_1 and β_2 thus capture the intent-to-treat effects of interest.

Table 2 presents the treatment effects from Equation 1. The first three columns reports the counts of the different parliamentary activities, and the last column reports results for an index created by equally weighting the three first three measures. Across each of the four outcomes, we fail to reject the null hypothesis of no effect. The finding is consistent across all individual measures of performance as well as the combined index. While the point estimates for both treatment arms remain negative, they are statistically indistinguishable from zero across the four specifications. Table A6 shows that the substantive conclusions remain unchanged, if we combine the two treatments into a binary treatment indicating a short term.

These null results alone are not sufficient to conclude that a shorter term of two or four years did not meaningfully alter members' performance. In order to rule out erroneous non-significant results, we investigate a number of alternate explanations (Kane 2025). By ruling out factors that could undermine the experimental setup, we conclude that the non-significant result represents a true absence of an effect. In the following paragraphs, we provide additional sets of evidence verifying the treatment compliance, assessment of power, formal equivalence tests and analysis of underlying heterogeneity.

First, results in Table 2 might be on account of the treatment not functioning as intended, due to non-compliance. This could happen if most members did not actually serve their assigned two- or four-year terms. We can empirically confirm that treatment manipulated the actual lengths of terms served. 192 out of the 214 members (about 90%) in the sample fully complied with the treatment they were assigned. For the 22 non-compliant members, 8 passed away during the term, 6 vacated their seats on being elected to the Lok Sabha and 8 resigned from office before the end of their terms. The high compliance and verification of valid reasons driving non-compliance confirm that the treatments functioned as intended.

Second, even with strict compliance and an effective treatment, a small sample may lack the statistical power required to detect a true effect, resulting in null findings. While we cannot control the sample sizes, our empirical strategy takes advantage of a number of features of the lottery that improve statistical power. Particularly, we take advantage of a blocking that makes sure experimental groups are balanced across states (Dolan 2023; Bailey 2016) and utilize both pre- and post-treatment measures of our outcomes (Clifford et al. 2021). Nevertheless, we employ power calculations to provide some information on the size of the effects that can be recovered from this experimental design. Power calculations in related research on term lengths indicates that statistical significance can be achieved due to large effect sizes in small samples (Titiunik 2016). We start by reporting

Table 2: Shorter terms do not worsen Parliamentary Performance

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
2 Year Term	−0.99	−1.82	−0.16	−0.07
	(5.20)	(4.93)	(0.19)	(0.09)
4 Year Term	−3.70	−0.64	−0.25	−0.08
	(5.80)	(3.08)	(0.18)	(0.08)
(B) Question Hour	8.34***			
	(1.00)			
(B) Written Questions		13.11**		
		(5.12)		
(B) Discussions			2.99**	
			(1.01)	
(B) Index				0.78***
				(0.13)
Control Means	28.96	10.22	0.6	0.1
State Fixed Effects	Yes	Yes	Yes	Yes
Observations	214	214	214	214
R^2	0.69	0.57	0.22	0.60

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of shorter terms on measures of legislative performance. "2 Years" and "4 Years" indicates a two-year and four-year term respectively. (B) denotes baseline period, that measures pre-treatment activity for all outcomes. All outcomes are count measures of the legislative activity. Controls include baseline levels of performance as shown in table.

the probability of detecting effects of different sizes in our set up. For each outcome, we calculate the power as the probability of rejecting the null hypothesis of no effect given that the alternate hypothesis is true.

We keep the experimental design fixed and use simulations of the data generating process[11] to test a range of hypothetical effect sizes. The original model specification—incorporating state fixed effects and robust standard errors—was re-estimated 1000 times per effect size, calculating statistical power as the proportion of iterations achieving significance ($p < 0.05$).

Figure 2 shows that see that the experiment had greater than 80% power to detect effect sizes between 0.25 and 0.4 SD across the four main outcomes and the two treatments: a setup sufficient to capture small sized effects. Related work on term lengths reports medium to large effect sizes on legislative performance (Titiunik 2016; Dal Bo and Rossi 2011), which this design was more than adequately powered to detect.

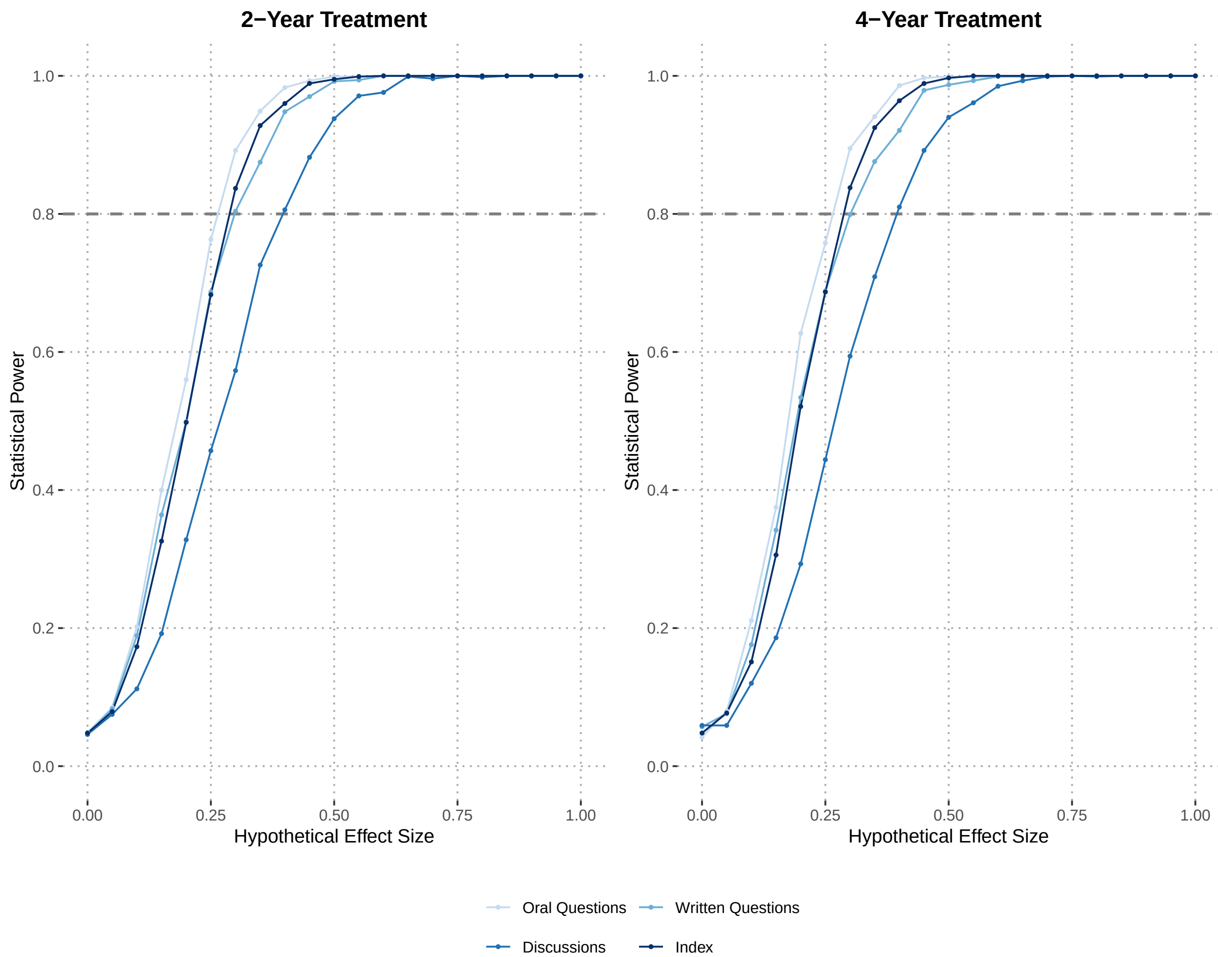
Formally, we conduct equivalence tests using two one-sided tests (TOST), which explicitly test whether meaningful effects are plausible in this setting (Rainey 2014). The equivalence tests flip the idea of the null hypothesis to ask whether the design of the experiment can recover the smallest effect size that is of substantive interest to researchers. Under this framework, we can reject a small effect size of 0.2 standard deviations or larger for participation in question hour and submitting written questions under the two-year treatment—the stronger of the two conditions. By ruling out small effects, these tests confirm that our findings point to a genuine null effect.

Third, we can dismiss poor measurement as drivers of these results; our primary outcomes rely on highly granular administrative records, and baseline participation rates indicate that members have ample room to further decrease their participation. A related concern is that parliamentary performance is a low-stakes metric that does not change with perverse incentives. To address this, Table A9 examines the effects of shorter terms on broader career trajectories and find no effects, reinforcing the validity of our primary findings.

Fourth, we address the concern that average treatment effects could be masking underlying heterogeneity. Treatment effects could occur in opposite directions for different subgroups: that is,

[11] Each simulation added a standardized effect into the treated units and added re-sampled residuals with replacement to preserve empirical noise.

Figure 2: Power analysis shows sufficient power to detect small effects between 0.25-0.4 SDs



Note: Power analysis conducted for each outcome separately for the two treatment groups. The X-axis denotes effect sizes from 0 to 1 times the standard deviation. The Y-axis plots power from 0 to 100%. We have 80% power to detect small effect sizes (about 0.25 times the SD or greater). This is a lower effect size that can be tested than that possible in previous work.

the positive treatment effects for one sub-group of members are washing out the negative treatment effects for another sub-group of members, which would be in line with the estimates in Table 2. Characteristics like the gender, education and previous experience of members could individually or together moderate the treatment effects.

In order to explore sources of heterogeneity, we leverage a new Bayesian semi-parametric method for fitting varying coefficient models called VCBART (Deshpande et al. 2024). We use VCBART to reanalyze data from our experimental set-up to overcome strict model assumptions of interaction terms to test heterogeneity (Bhogale and Deshpande 2025). In the VCBART set up, we model our outcome index in the following varying-coefficient set-up:

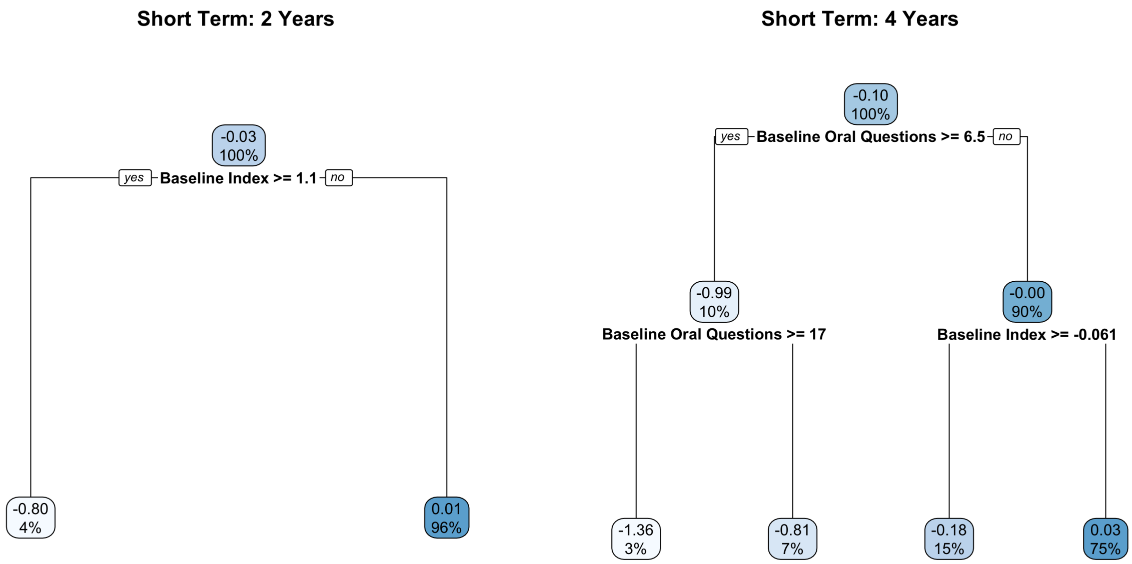
$$Y = \beta_0(\mathbf{X}) + \beta_1(\mathbf{X})T1 + \beta_2(\mathbf{X})T2 + \epsilon \tag{2}$$

wherein $\beta_0(\mathbf{X})$ are functions of the pre-treatment characteristics $\mathbf{X}$ for each member, which help us construct the conditional average treatment effects, or CATEs. We then fit a single classification and regression tree to identify the main drivers of CATEs and group members into relevant subgroups. Figure 3 indicates that only the pre-treatment measures of performance meaningfully explain the variation in treatment effects. In other words, how a member responds to a the short term treatments depends exclusively on their baseline level of activity, rather than their demographic or political characteristics.

Finally, absence of an overall treatment effect could also stem from aggregating outcomes over a two year period. A theoretical channel through which shorter terms worsen performance is election proximity. As the time to an election decreases, the incentive to participate in parliamentary activities may weaken, since MPs redirect their time toward campaigning. If this were true, high initial performance could mask a subsequent decline in activity as elections come closer.

To demonstrate that this is not the case, we leverage the fine-grained daily data on parliamentary activities. To test whether shorter terms induce end-of-term shirking, we calculate the mean values for our outcomes over each session and estimate the treatment effect for each session by interacting our two treatments with a categorical indicator denoting separate sessions.

Figure 3: Heterogeneity is only driven by baseline activity



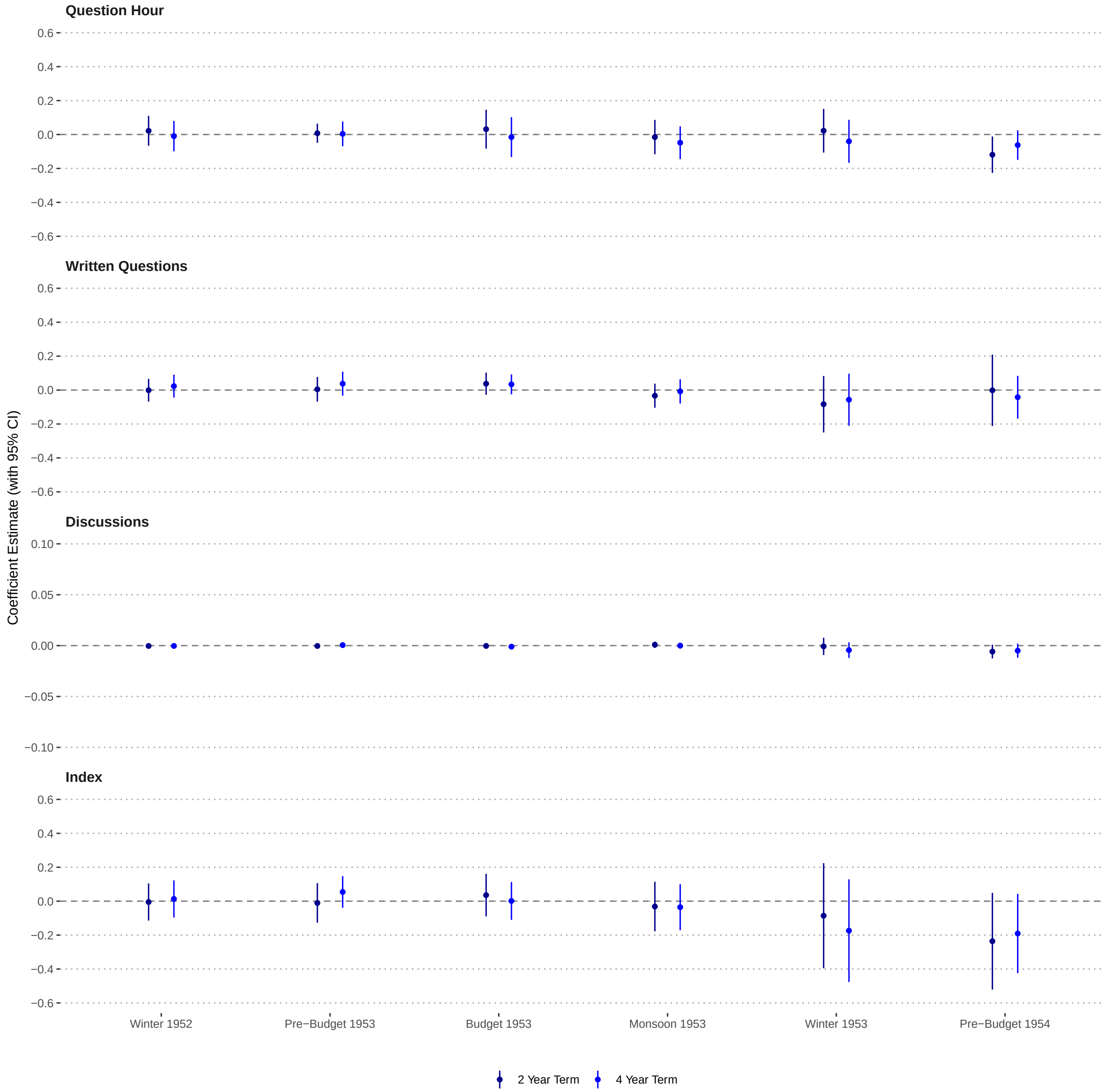
Note: Outcome is an index of legislative performance created by taking the scaled means of the five main outcomes. Treatment effects (-0.03, -0.00) measured in SDs of the index.

In Figure 4 we see no temporal trends in the treatment effects. All estimates remain statistically insignificant at conventional levels. A single exception is a marginal decrease in question hour activity during the 1954 pre-Budget Session for the 2-year group, however, this isolated result lacks a systematic trend and is not mirrored in any other outcomes. We replicate this exercise for the 4-Year and 6-Year groups that remain in the house between 1954 and 1956 in Figure A1, which lends further confidence to the absence of an election proximity effect for these MPs. This confirms that the overall null effect remains consistent throughout the electoral cycle.

6 Selection drives effects

So far, the results show that the high quality of members that were a part of the founding cohort of the *Rajya Sabha* remain unresponsive to the negative incentives created by shorter terms. However, the absence of a treatment effect in Section 5 could imply two things: the first implication, in line with our theory, is that the positive selection of members overcomes negative incentives. An alternate explanation is that the nature of indirect elections entirely insulates representatives from

Figure 4: Proximity to elections does not modify behavior



Note: Each outcome is a separate panel, displaying 2-year term and 4-year term coefficients for each parliamentary session separately. Full regression results shown in Table A7.

electoral pressure, removing the negative incentives. This might indeed be the case, given that Table A5 indicates that the average elected member in the Rajya Sabha is not particularly different from the founding members elected in 1952.

In this section, we discuss whether the indirect mode of election is sufficient to guarantee performance, and find that shorter terms can worsen parliamentary performance once positive selection is not guaranteed. We show this by leveraging a different research design within the same indirectly-elected house. Here, we study the effect of shorter terms on legislators who are indirectly elected during by-elections over a 35-year period between 1960 and 1995. Mid-term by-elections are regularly conducted to fill vacancies on account of a sitting member's death, resignation, or disqualification. While such members are indirectly elected by the intermediary electors, Table A4 indicates that they no longer have the higher quality that was typical of members elected in 1952. Because by-elections happen between regular election cycles, most top political candidates already hold other state or national roles, leaving a much smaller pool of available candidates. Under these circumstances, selection of members of the highest quality is unlikely. While shorter terms may worsen the performance of these members, the successive filters of indirect-elections may be sufficient to counteract the negative effects.

To test this idea, we leverage the exogenous timing of an incumbent *Rajya Sabha* member's death to estimate the effect of shorter term lengths on parliamentary performance. Typically, a member of the *Rajya Sabha* continues to hold their position for six years, unless the seat is vacated due to resignation or death. This design uses the death of a sitting member as an unanticipated shock that leads to an exogenous assignment of different lengths of remaining terms to the replacement MPs. For instance, if an MP passes away in the first year when they are holding office, this event assigns a five-year term for the replacement MP, and similarly, if an MP dies in the fifth year of holding office, a replacement MP is elected to serve for one year.

While the design provides a significant methodological advantage over purely observable studies and has been leveraged before in different contexts (Fong 2020; Baskaran et al. 2015), we start by assessing its validity in the *Rajya Sabha*. We identify the members of the Rajya Sabha who passed away mid-session between 1960 and 1995, an event which prompted a by-election to fill the vacant seat.[12] We use data on members' electoral terms[13] identify all deaths and the replacement MPs elected.[14]

We start by comparing the backgrounds of the replacement MPs to those elected in 1952. We find evidence that replacement MPs tend to be of lower quality than those elected in 1952. Table A4 shows that the two groups differ across several background characteristics that serve as proxies for politician quality. While similar in age, the MPs elected in 1952 possess, on average, more legislative experience, higher levels of education, and are more likely to come from a legal background. Importantly, while a number of members in 1952 served in the provincial parliament and as a drafting member of the constitution, none of the replacement MPs held these positions.

We also use the member characteristics to conduct covariate balance tests to demonstrate that the observable pre-treatment characteristics of the replacement MPs are orthogonal to the assigned term lengths. Parties could strategically sorting 'high-quality' talent into longer terms, and if this were the case, we would expect to see a consistent correlation between background characteristics and assigned term lengths. For the purpose of these balance tests, we define that a member is a 'Short Term Replacement MP' if the member is assigned less than four years as their term length, to keep the definition consistent with the first research design.

Figure 5 shows that the replacement MPs assigned shorter and longer terms come from similar backgrounds. Shorter term replacement MPs are no more likely to have a high school education or a graduate degree. Against the expectation that longer terms would attract a higher quality of MPs, we observe that MPs assigned shorter terms are slightly more likely to hold a post-graduate degree. We further rule out any differences in backgrounds out by conducting a joint F-test: this yields an F-statistic of 1.24 ($p = 0.28$), failing to reject the null hypothesis that the background characteristics are jointly unable to predict treatment assignment.[15] We conclude that the variation in term-lengths

[12]While a by-election is typically conducted when a sitting MP dies, it can also happen due to an MP switching to another party, or rarely in cases of mergers or factions of parties in the House. We only consider the by-elections for members for whom a term is assigned due to the death of a sitting MP.

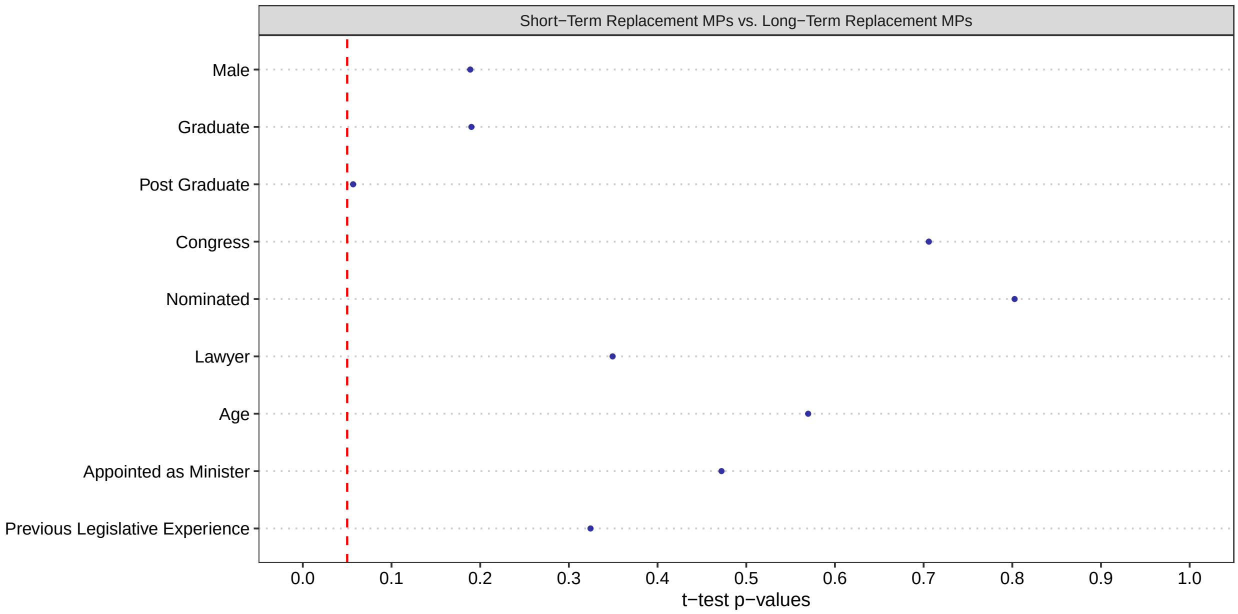
[13]*Rajya Sabha Members' Dataset* from the Trivedi Center for Political Data.

[14]According to the official procedures of the *Rajya Sabha*, a by-election within six months is required to be conducted. We use this heuristic to match the replacement MPs to those who vacated their seats due to their death. However, when the remaining term is less than one year, then the by-election is not required, as per procedures of the *Rajya Sabha* explained in *Rajya Sabha at Work*, which can be accessed at: https://rajyasabha.nic.in/Procedures/RajyasabhaAtWork.

[15]While the pool of candidates available to fill these vacant positions might shift due to the attractiveness of the remaining term, we are able to show that the *winners* to these seats do not systematically differ by quality. Further, the

induced by the random timing of death results in a balanced group of replacement MPs.

Figure 5: Replacement MPs are balanced across treatment and control groups



Note: Dotted red line indicates the 0.05 threshold for the p-values from t-tests. Variables are indicators, except *Age*. All replacement MPs were not a part of the constituent assembly and provincial parliament. Table A3 displays exact group means and p-values.

While Figure 5 shows that members are similar across *measured* characteristics, we are concerned about *unobserved* variables that may remain imbalanced and related to our outcome of interest. For example, parties could select party loyalists if the length of the remaining term is a single year. Parties could select members with greater motivation if the length of the remaining term is over five years. In addition to the party-level selection, the candidate pool itself can be shaped by the attractiveness of the remaining duration: those with policy-making expertise may only elect to run for longer terms.

If these dynamics are true, they would introduce selection bias, wherein the observed differences in performance could reflect the quality of the replacement members. To account for this—and although a natural experiment framework is plausible—we employ a selection-on-observables design, and supplement our results with sensitivity analysis to assess the impact of potential omitted

elections are conducted via an open ballot and leaders are frequently elected unopposed. For instance, in the 2026 Rajya Sabha elections, over 70% of representatives were elected unopposed.

variable bias on our conclusions (Cinelli and Hazlett 2020) along with robustness to choice of estimation method (Quinn et al. 2026).

6.1 Shorter terms worsen performance of Replacement MPs

We focus this analysis on the Replacement MPs: who are elected at times when intermediary electors may not have the opportunity to select high-quality members to the *Rajya Sabha*. To test this, leverage variation in term lengths following the death of a sitting member: a replacement MP can serve as little as 4 months all the way upto 5.5 years. For the estimation, S_i is an indicator coded one if MP i served a remaining term of four years or less, and zero if i when the remaining term was more than four years. X_i is a vector of observed covariates for the replacement MPs. The outcome Yi measures the same four outcomes studied in the previous section.

We proceed with the selection on observables framework, by assuming unconfoundedness and common support. Our rich set of covariates includes meaningful background characteristics that could be used to manipulate the assignment mechanisms—conditioning on **X** the treatment assignment is as good as random and eliminates selection bias.

To estimate how a shorter term affects parliamentary performance, we measure each replacement MP's performance as a mean of their daily activities for each days that the *Rajya Sabha* convened when they served. We use OLS regression for covariate adjustment using the following specification:

$$Y_i = \alpha + \beta_1 S_i + \gamma \mathbf{X}_i + \delta_{y(i)} + \epsilon_i \tag{3}$$

Where Y_i is the average of the daily outcomes for MP i, S_i denoting that the MP served a shorter term: that is, MP i served four or fewer than four years upon taking office. $\mathbf{X}_i$ is a vector of time-invariant MP-level controls, which includes gender, education, party background, an indicator for whether the MP is a nominated member or minister, age and years of prior legislative experience. $\delta_{y(i)}$ represents start-year fixed-effects, which absorbs all political shocks in the decade in which the MP took office. ϵ_i is the error term.

Table 3 presents the main results for the four outcomes with robust standard errors. We see that shorter terms for this group of MPs worsened performance across all outcomes, each of which is statistically significant. The point estimates indicate that the being assigned a shorter term reduces the MP's daily participation in activities by between 64-75%. Assuming that the *Rajya Sabha* met and conducted activities over 90 days in a year, these estimates show that the average MP serving a shorter term raised 6.3 fewer discussions during the question hour, submitted 90 fewer written questions and participated in 1.8 fewer discussions. For all three outcomes and the combined index, we reject the null hypothesis of no effect at conventional levels.

A natural concern with this analysis is that it compares MPs in different stages of their term. An MP serving a 1-year term is only observed during the last year of their tenure, while an MP serving a 5-year term is observed over 5 years, when they can be many years away from the end of their term and behave differently. In Table A12, we show that the result remain robust to focusing on activities during the replacement MPs' last two years in office. We find that the estimates remain both statistically significant and similar in magnitude to the findings in Table 3.

To assess whether our estimates are vulnerable to unobserved confounding, we conduct sensitivity analyses based on the framework in Cinelli and Hazlett (2020), focusing on the index of parliamentary performance. The sensitivity analysis reveals that an unobserved confounder would have to explain atleast 30.5% of the residual variance in both the treatment and the outcome to fully account for the estimated effect. Such a confounder is unlikely, in that it would need to explain more than three times the variation explained by membership to the ruling Congress party. Benchmarking a hypothetical confounder against Congress party membership is meaningful, given that our sample spans the period between 1960 and 1995, decades that were largely characterized by Congress dominance at both the center and state levels. Thus, party affiliation with the Congress serves as a powerful proxy for institutional backing and political capital. Figure A2 displays the sensitivity contour plots for the estimates and t-values. These bounds confirm that the estimated effect of the length of term on the index remain robust; any unmeasured confounder would need to be much more influential than party affiliation with the ruling party to explain away the effect. In Table A10, we report estimates for different estimation strategies, based on the guidance following Quinn et al. (2026). We find that the results remain similarly robust, and that our results are not driven by the

Table 3: Shorter terms worsen performance of Replacement MPs

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
Short Term	-0.07^{**}	-0.45^{**}	-0.02^{**}	-0.69^{**}
	(0.03)	(0.19)	(0.01)	(0.27)
Minister	-0.04	-0.57^{**}	-0.04^{*}	-1.05^{*}
	(0.04)	(0.27)	(0.02)	(0.54)
Prior Experience	-0.00	-0.01	0.00	0.01
	(0.00)	(0.01)	(0.00)	(0.02)
Male	-0.02	0.07	0.01	0.17
	(0.03)	(0.19)	(0.01)	(0.27)
Post-Graduate	-0.00	0.25	0.02	0.45
	(0.03)	(0.22)	(0.01)	(0.34)
Graduate	0.03	0.07	0.00	0.11
	(0.04)	(0.21)	(0.01)	(0.27)
Congress Party	-0.01	-0.21	-0.01	-0.31
	(0.02)	(0.16)	(0.01)	(0.21)
Nominated Member	0.02	-0.55^{*}	0.04^{**}	-0.15
	(0.07)	(0.30)	(0.02)	(0.39)
Law Background	-0.02	-0.01	0.01	-0.05
	(0.03)	(0.23)	(0.01)	(0.31)
Control Means	0.11	0.69	0.04	0.59
Start-Decade Fixed Effects	Yes	Yes	Yes	Yes
Observations	77	77	77	77
R^2	0.14	0.25	0.33	0.27

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of length of term on measures of legislative performance for MPs elected post death of a sitting member.

choice of estimation methods. A final concern with this analysis is that it could be conflating term lengths with institutional capacity: in Table A13, we look at the effect of a shorter term on MPs who had previous experience as a legislator. While this considerably reduces the sample size, the results remain consistent with those presented in Table 3, both in magnitude and direction.

7 Discussion

Political scientists have long been interested in how the design of institutions shapes incentives and affects government effectiveness. Yet, it remains unclear whether these incentives operate uniformly regardless of the quality of actors. In this paper, we theorize and empirically demonstrate the conditions that moderate the effects of adverse incentives. Specifically, we argue that high-quality of representatives can overcome the pressures created by shorter term lengths, as the mechanisms driving these perverse incentives fail to take hold among high-quality representatives.

Leveraging evidence from an indirectly-elected legislature to report these patterns is consequential for two reasons. The experimental set-up provides a clean lottery that can be evaluated to study the effect of term lengths on members who are positively selected. Further, the selection-on-observables strategy leveraging variation in term lengths following deaths of sitting members provides an avenue to study the effect of term lengths in the same house and under the same institutional rules, once positive selection is no longer viable. The results show that the perverse incentives of shorter terms do not affect high quality members, and that the effects are driven by selection and not the general nature of an indirectly elected house.

We overcome the traditional difficulty of studying both selection and incentives by introducing two novel research designs, that can be applied across other comparative contexts. While previous work has leveraged randomized term lengths in Argentina (Dal Bo and Rossi 2011) and the United States (Titiunik 2016; Gaines et al. 2012; Pomirchy 2023), we exploit a lottery-assigned randomization that is highly generalizable to other settings. A direct extension of this work would be to evaluate performance in other legislatures with staggered entry and exit rules, such as Indian state upper houses, the French Senate, Kazakhstan's Senate, Pakistan's Senate, and Nepal's National Assembly. Complementing this, our second design estimates the impact of shorter term lengths by analyzing replacement MPs elected following the death of a sitting member—a strategy that can be easily adapted to study term lengths across nearly any global legislature.

A natural limitation of this setting is that we are only able to study performance in the legislature, as opposed to MPs' efforts in law-making. While the parliamentary devices studied in this paper allows us to study how often members exercise oversight and raise urgent matters, the introduction of bills and discussion on legislation is tightly controlled by party leaders and whips in the Indian legislatures, and is not indicative of individual effort. Future work can turn to other comparative legislatures to investigate the effect of incentives on legislative efforts, and particularly, how these patterns interact with the quality of representatives.

While institutional incentives can impact performance, researchers must carefully consider the underlying forms of political representation. Observed outcomes of performance not just shaped by institutional rules, but also by the mode of selection. Legislatures composed of indirectly elected members vary heavily in their makeup: for instance, while the Indian *Rajya Sabha* contains only a small fraction of President-nominated members, over a third of the seats in other indirectly elected houses are appointed by the head of state. Where selection mechanisms are highly centralized, the effect of incentives can be similarly muted—an area open for further exploration.

References

Bailey, Michael A. 2016. *Real Stats: Using Econometrics for Political Science and Public Policy*. 1st edition. Oxford University Press.

Baskaran, Thushyanthan, Brian Min, and Yogesh Uppal. 2015. "Election Cycles and Electricity Provision: Evidence from a Quasi-Experiment with Indian Special Elections." *Journal of Public Economics* 126 (June): 64–73. https://doi.org/10.1016/j.jpubeco.2015.03.011.

Bernhard, William, and Brian R. Sala. 2006. "The Remaking of an American Senate: The 17th Amendment and Ideological Responsiveness." *The Journal of Politics* 68 (2): 345–57. https://doi.org/10.1111/j.1468-2508.2006.00411.x.

Bhatia, Udit. 2022. "Indirect Elections as a Constitutional Device of Epistocracy." *International Journal of Constitutional Law* 20 (1): 82–111. https://doi.org/10.1093/icon/moac001.

Bhogale, Saloni, and Sameer K. Deshpande. 2025. "Estimating Treatment Effect Heterogeneity with Varying Coefficients and Bayesian Tree Ensembles." Unpublished manuscript.

Callen, Michael, Saad Gulzar, Ali Hasanain, Muhammad Yasir Khan, and Arman Rezaee. 2025. "Personalities and Public Sector Performance: Evidence from a Health Experiment in Pakistan." *Economic Development and Cultural Change* 73 (3): 1439–74. https://doi.org/10.1086/731673.

Chaudhuri, Ananish, Vegard Iversen, Francesca R. Jensenius, and Pushkar Maitra. 2025. "Selecting the 'Best'? Competing Dimensions of Politician Quality in the Developing World." *British Journal of Political Science* 55: e81. https://doi.org/10.1017/S0007123425000225.

Cinelli, Carlos, and Chad Hazlett. 2020. "Making Sense of Sensitivity: Extending Omitted Variable Bias." *Journal of the Royal Statistical Society Series B: Statistical Methodology* 82 (1): 39–67. https://doi.org/10.1111/rssb.12348.

Clifford, Scott, Geoffrey Sheagley, and Spencer Piston. 2021. "Increasing Precision Without Altering Treatment Effects: Repeated Measures Designs in Survey Experiments." *American Political Science Review* 115 (3): 1048–65. https://doi.org/10.1017/S0003055421000241.

Dal Bo, E., and M. A. Rossi. 2011. "Term Length and the Effort of Politicians." *The Review of Economic Studies* 78 (4): 1237–63. https://doi.org/10.1093/restud/rdr010.

Deshpande, Sameer K., Ray Bai, Cecilia Balocchi, Jennifer E. Starling, and Jordan Weiss. 2024. "VCBART: Bayesian Trees for Varying Coefficients." *Bayesian Analysis* -1 (-1). https://doi.org/10.1214/24-BA1470.

Dolan, Lindsay. 2023. "10 Things to Know About Randomization." In *EGAP*.

Election Commission of India. 1955. *Report on the First General Elections in India 1951-52*. Volume 1 (General).

Fagan, E. J., and Zachary A. McGee. 2020. "Problem Solving and the Demand for Expert Information in Congress." *Legislative Studies Quarterly*, December, lsq.12323. https://doi.org/10.1111/lsq.12323.

Ferraz, Claudio, and Frederico Finan. 2009. *Motivating Politicians: The Impacts of Monetary Incentives on Quality and Performance*. No. w14906. National Bureau of Economic Research. https://doi.org/10.3386/w14906.

Fong, Christian. 2020. "Expertise, Networks, and Interpersonal Influence in Congress." *The Journal of Politics* 82 (1): 269–84. https://doi.org/10.1086/705816.

Fukumoto, Kentaro, and Akitaka Matsuo. 2015. "The Effects of Election Proximity on Participatory Shirking: The Staggered-Term Chamber as a Laboratory." *Legislative Studies Quarterly* 40 (4): 599–625. https://doi.org/10.1111/lsq.12090.

Gailmard, Sean, and Jeffery A. Jenkins. 2009. "Agency Problems, the 17th Amendment, and Representation in the Senate." *American Journal of Political Science* 53 (2): 324–42. https://doi.org/10.1111/j.1540-5907.2009.00373.x.

Gaines, Brian J., Timothy P. Nokken, and Collin Groebe. 2012. "Is Four Twice as Nice as Two? A Natural Experiment on Electoral Effects of Term Length." *State Politics & Policy Quarterly* 12 (1): 43–57. https://doi.org/10.1177/1532440011433588.

Galasso, Vincenzo, and Tommaso Nannicini. 2011. “Competing on Good Politicians.” *American Political Science Review* 105 (1): 79–99. https://doi.org/10.1017/S0003055410000535.

Gersbach, Hans, Matthew O. Jackson, and Oriol Tejada. 2020. “The Optimal Length of Political Terms.” *SSRN Electronic Journal*, ahead of print. https://doi.org/10.2139/ssrn.3615407.

Grose, Christian R. 2021. “Experiments, Political Elites, and Political Institutions.” In *Advances in Experimental Political Science*, 1st ed., edited by James Druckman and Donald P. Green. Cambridge University Press. https://doi.org/10.1017/9781108777919.011.

Grumbach, Jacob M., and Alexander Sahn. 2020. “Race and Representation in Campaign Finance.” *American Political Science Review* 114 (1): 206–21. https://doi.org/10.1017/S0003055419000637.

Gulzar, Saad. 2021. “Who Enters Politics and Why?” *Annual Review of Political Science* 24 (1): 253–75. https://doi.org/10.1146/annurev-polisci-051418-051214.

Gulzar, Saad, and Muhammad Yasir Khan. 2025. “Good Politicians: Experimental Evidence on Motivations for Political Candidacy and Government Performance.” *Review of Economic Studies* 92 (1): 339–64. https://doi.org/10.1093/restud/rdae026.

Heinze, Alyssa R. 2025. “Democratic Deepening or Elite Persistence? How Local Elites Adapt to Electoral Reform in Rural India.” *American Political Science Review*, October, 1–20. https://doi.org/10.1017/S0003055425101068.

Kane, John V. 2025. “More Than Meets the ITT: A Guide for Anticipating and Investigating Nonsignificant Results in Survey Experiments.” *Journal of Experimental Political Science* 12 (1): 110–25. https://doi.org/10.1017/XPS.2024.1.

Kellermann, Michael. 2009. “Congressional Careers, Committee Assignments, and Seniority Randomization in the US House of Representatives.” *Quarterly Journal of Political Science* 4 (2): 87–101. https://doi.org/10.1561/100.00008061.

Mansbridge, Jane. 2009. “A ‘Selection Model’ of Political Representation*.” *Journal of Political Philosophy* 17 (4): 369–98. https://doi.org/10.1111/j.1467-9760.2009.00337.x.

Micozzi, Juan Pablo. 2013. "Does Electoral Accountability Make a Difference? Direct Elections, Career Ambition, and Legislative Performance in the Argentine Senate." *The Journal of Politics* 75 (1): 137–49. https://doi.org/10.1017/S0022381612000928.

Napolio, Nicholas G., and Christian R. Grose. 2021. "Crossing Over: Majority Party Control Affects Legislator Behavior and the Agenda." *American Political Science Review*, August, 1–8. https://doi.org/10.1017/S0003055421000721.

Pomirchy, Michael. 2023. "Electoral Proximity and Issue-Specific Responsiveness." *Public Opinion Quarterly* 87 (3): 662–88. https://doi.org/10.1093/poq/nfad031.

Quinn, Kevin M., Guoer Liu, Lee Epstein, and Andrew D. Martin. 2026. "What to Observe When Assuming Selection on Observables." *Political Analysis* 34 (1): 1–22. https://doi.org/10.1017/pan.2025.10003.

Rainey, Carlisle. 2014. "Arguing for a Negligible Effect." *American Journal of Political Science* 58 (4): 1083–91. https://doi.org/10.1111/ajps.12102.

Rogers, Steven. 2012. "The Responsiveness of Direct and Indirect Elections." *Legislative Studies Quarterly* 37 (4): 509–32. https://doi.org/10.1111/j.1939-9162.2012.00060.x.

Schultz, Christian. 2008. "Information, Polarization and Term Length in Democracy." *Journal of Public Economics* 92 (5-6): 1078–91. https://doi.org/10.1016/j.jpubeco.2007.12.008.

Titiunik, Rocío. 2016. "Drawing Your Senator from a Jar:term Length and Legislative Behavior." *Political Science Research and Methods* 4 (2): 293–316. https://doi.org/10.1017/psrm.2015.20.

Zelizer, Adam. 2019. "Is Position-Taking Contagious? Evidence of Cue-Taking from Two Field Experiments in a State Legislature." *American Political Science Review* 113 (2): 340–52. https://doi.org/10.1017/S0003055419000078.

Appendix

A Legislatures with Indirect Elections

Table A1 shows countries with indirectly elected members in national legislatures from the Inter-Parliamentary Union (IPU). While directly-elected houses are more common across countries, we observe the prevalence of indirectly-elected houses, predominantly as a part of a bicameral legislature, across countries across the Global North and South.

Table A1: Legislatures whose members are indirectly elected as of 2025

Code	Country	Chamber	Structure	Term Length
NL	Netherlands	Senate	Bicameral	4
BY	Belarus	Council of the Republic	Bicameral	5
BE	Belgium	Senate	Bicameral	5
BI	Burundi	Senate	Bicameral	5
CM	Cameroon	Senate	Bicameral	5
CI	Côte d'Ivoire	Senate	Bicameral	5
CD	Congo (DRC)	Senate	Bicameral	5
ET	Ethiopia	House of the Federation	Bicameral	5
GA	Gabon	Senate	Bicameral	5
IE	Ireland	Senate	Bicameral	5
MG	Madagascar	Senate	Bicameral	5
NA	Namibia	National Council	Bicameral	5
RW	Rwanda	Senate	Bicameral	5
SI	Slovenia	National Council	Bicameral	5
SO	Somalia	House of the People	Bicameral	5
SO	Somalia	Upper House	Bicameral	5
TJ	Tajikistan	National Assembly	Bicameral	5
TH	Thailand	Senate	Bicameral	5
TN	Tunisia	National Council of Regions and Districts	Bicameral	5
UZ	Uzbekistan	Senate	Bicameral	5
AT	Austria	Federal Council	Bicameral	5,6
DZ	Algeria	Council of the Nation	Bicameral	6
KH	Cambodia	Senate	Bicameral	6
TD	Chad	Senate	Bicameral	6
CG	Congo	Senate	Bicameral	6
FR	France	Senate	Bicameral	6
IN	India	Council of States	Bicameral	6
KZ	Kazakhstan	Senate	Bicameral	6
MA	Morocco	House of Councillors	Bicameral	6
NP	Nepal	National Assembly	Bicameral	6
PK	Pakistan	Senate	Bicameral	6
TG	Togo	Senate	Bicameral	6
CN	China	National People's Congress	Unicameral	5

Note: Data coded from Inter-Parliamentary Union's Parline dataset

Table A2: Background characteristics balanced between treatments and control group

Variable	$Mean_C$	$Mean_{T_1}$	$Mean_{T_2}$	$p.val_{C,T_1}$	$p.val_{C,T_2}$
Male	0.93	0.96	0.90	0.48	0.53
High School	0.12	0.14	0.13	0.78	0.97
Graduate	0.46	0.49	0.46	0.68	0.94
Post Graduate	0.28	0.34	0.35	0.44	0.34
Congress	0.68	0.69	0.75	0.9	0.39
Independent	0.06	0.01	0.06	0.18	0.98
Nominated	0.06	0.06	0.06	0.98	0.98
Lawyer	0.36	0.37	0.34	0.95	0.77
Age	50.07	51.20	50.35	0.57	0.89
Appointed as Minister	0.03	0.06	0.03	0.4	0.99
Member of Constituent Assembly	0.15	0.07	0.17	0.12	0.79
Member of Provincial Parliament	0.17	0.11	0.18	0.36	0.8
Previous Legislative Experience	8.66	7.74	10.85	0.58	0.28
Baseline Question Hour	2.61	2.30	1.86	0.76	0.42
Baseline Written Questions	0.94	0.75	0.48	0.66	0.28
Baseline Discussion	0.03	0.04	0.00	0.7	
Baseline Index	0.06	0.04	-0.10	0.9	0.15

Note: Comparison of the background of members who are assigned the two short-term treatments T1 or T2 (two and four years respectively) and the members in the control group who are assigned the regular six-year term. All variables are binary, except for age and previous legislative experience, which are measured in years. First three columns display the mean values for each group, and the next three columns display the p-values from the t-test. The large p-values from the t-test demonstrate that the members assigned the two shorter-term treatments are similar to those assigned longer terms.

B Balance Tests

B.1 Balance test on background characteristics

In this section, we reproduce the balance tests comparing the three groups of MPs in separate t-tests. We find that different groups are balanced across demographic characteristics, as seen in Table A2 for the MPs in the 1952 lottery and in Table A3 for the replacement MPs that follow deaths of sitting members. We see that the coefficients are balanced across all covariates with p-values well over 0.05.

B.2 Comparing Representatives in the Rajya Sabha

In Table A4, we compare the characteristics of MPs elected in 1952 with the replacement MPs analysed in the second part of the paper.

An important question is that which group may be more representative of MPs in general in the Rajya

Table A3: Background characteristics balanced between treatments and control group

Variable	$Mean_C$	$Mean_T$	$p.val_{C,T}$
Male	0.73	0.90	0.19
High School	0.00	0.00	
Graduate	0.07	0.18	0.19
Post Graduate	0.47	0.18	0.06
Congress	0.67	0.61	0.71
Independent	0.00	0.00	
Nominated	0.07	0.05	0.8
Lawyer	0.27	0.15	0.35
Age	28.40	31.47	0.57
Appointed as Minister	0.07	0.02	0.47
Previous Legislative Experience	5.57	3.35	0.32

Note: Comparison of the background of replacement MPs who are assigned the treatments T (fewer than four years remaining) and the members in the control group who are assigned a term that lasts more than four years. All variables are binary, except for age and previous legislative experience, which are measured in years. First two columns display the mean values for each group, and the next two columns display the p-values from the t-test. The large p-values from the t-test demonstrate that the members assigned the shorter-term are similar to those assigned longer terms.

Table A4: Comparing the two samples

Variable	MPs in 1952	Replacement MPs	p-val(t-test)
Age	50.54	51.84	0.45
Law background	0.36	0.17	0.00
Years of Legislative Experience	9.17	3.78	0.00
Graduate	0.47	0.16	0.00
Post Graduate	0.32	0.23	0.13
Member of Provincial Parliament	0.15	0.00	0.00
Member of Constituent Assembly	0.13	0.00	0.00

Table A5: Comparing MPs in the Rajya Sabha

Variable	$Mean_{All}$	$Mean_{1952}$	$Mean_{RepMP}$	$p.val_{All,1952}$	$p.val_{All,RepMp}$
Male	0.91	0.93	0.87	0.23	0.28
Age	52.55	50.54	54.09	0.02	0.34
Graduate	0.32	0.47	0.16	0.00	0.00
Post Graduate	0.29	0.32	0.23	0.29	0.29
Law Background	0.33	0.32	0.17	0.8	0.00

Sabha. In Table A5, we compare a subset of these characteristics with all MPs between 1952 and 1995. $Mean_{All}$ includes members who were elected during regular electoral cycles between 1954 and 1995. $Mean_{1952}$ includes the 214 members whose term lengths were assigned using a lottery. $Mean_{RepMP}$ includes the "Replacement MPs" who were elected after the death of a sitting member to serve the remainder of the term. We find that members in 1952 have a larger proportion of graduates as compared to the members elected during the regular election cycles, while the replacement MPs are less likely to hold a graduate degree and less likely to come from a law background relative to both groups.

C Additional Results: Lottery Experiment

C.1 Binary Treatment Definition

In this section, we show robustness to alternate definition of the treatment, by estimating the effects of a shorter term by combining the 2-year and 4-year treatment. We see that the results remain robust from Table A6.

C.2 Session-Wise Results

In Table A7 we produce the regression results for Figure 7 from the paper. In Figure A1 we produce the results comparing the 4-Year and 6-Year MPs after the 2-Year group has resigned, in the period between April 1954 and April 1956. Table A8 displays the associated regression table.

Table A6: Results robust to binary treatment

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
Short Term	−2.34	−1.23	−0.20	−0.08
	(4.78)	(3.55)	(0.18)	(0.08)
(B) Question Hour	8.35***			
	(1.00)			
(B) Written Questions		13.10**		
		(5.11)		
(B) Discussions			3.01**	
			(1.01)	
(B) Index				0.78***
				(0.13)
Control Means	28.96	10.22	0.6	0.1
State Fixed Effects	Yes	Yes	Yes	Yes
Observations	214	214	214	214
R^2	0.68	0.57	0.22	0.60

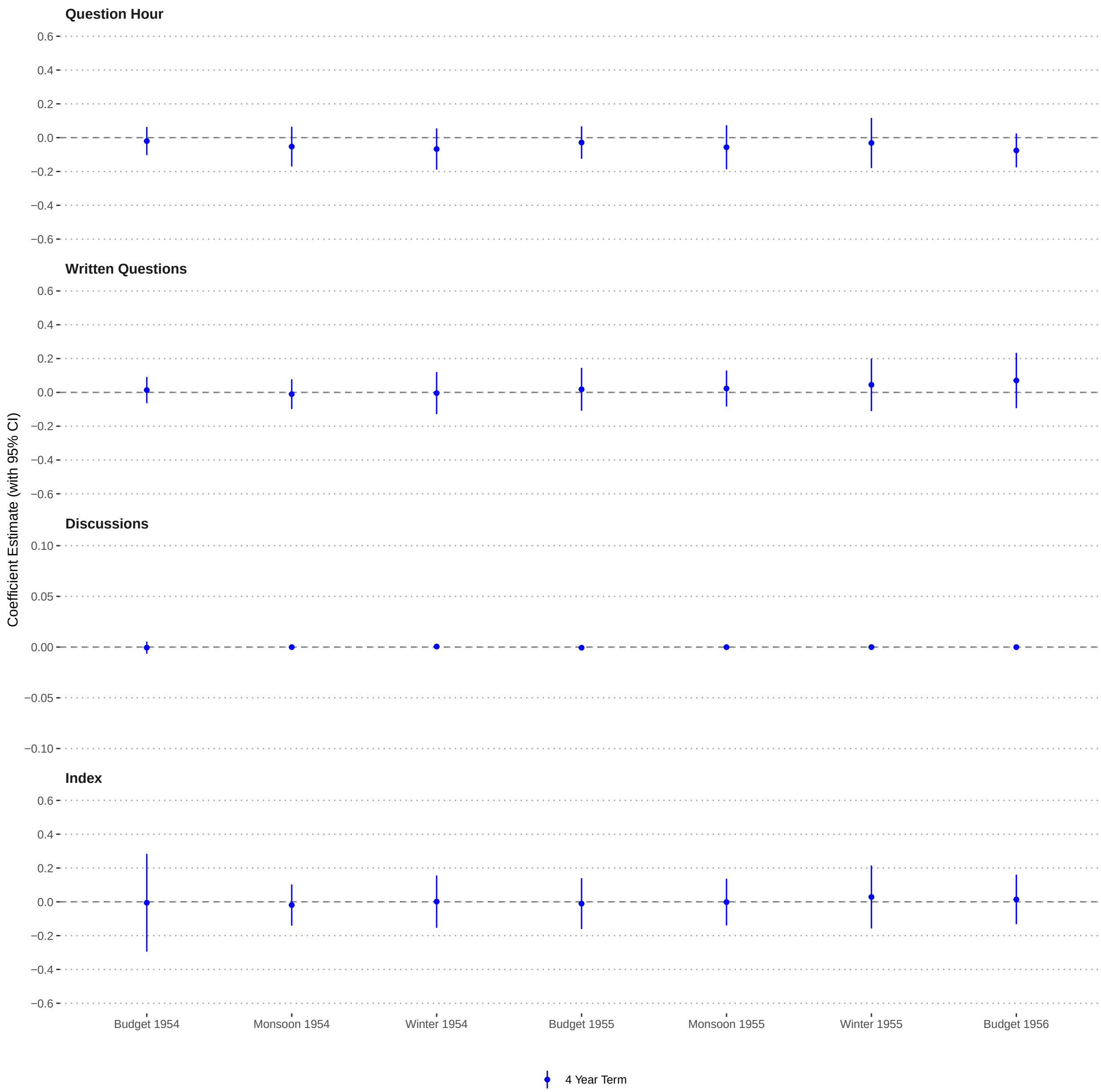
Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of shorter terms on measures of legislative performance. 'Short-Term' indicates a two-year or a four-year term. (B) denotes baseline period, that measures pre-treatment activity for all outcomes. All outcomes are count measures of the legislative activity. Controls include baseline levels of performance as shown in table.

Table A7: Term Lengths and Legislative Performance by Session

	Question Hour	Written Questions	Discussions	Index
Pre-Budget 1953	−0.06	−0.02	−0.00	−0.08
	(0.04)	(0.04)	(0.00)	(0.06)
Budget 1953	0.03	−0.01	0.00	0.03
	(0.05)	(0.04)	(0.00)	(0.06)
Monsoon 1953	0.03	0.05	0.00	0.10
	(0.05)	(0.04)	(0.00)	(0.07)
Winter 1953	0.03	0.10	0.01^{***}	0.33^{**}
	(0.06)	(0.09)	(0.00)	(0.15)
Pre-Budget 1954	0.03	0.08	0.01^{**}	0.27^{**}
	(0.04)	(0.08)	(0.00)	(0.12)
Winter 1952 × 2 Year	0.02	−0.00	−0.00	−0.00
	(0.04)	(0.03)	(0.00)	(0.06)
Pre-Budget 1953 × 2 Year	0.01	0.00	−0.00	−0.01
	(0.03)	(0.04)	(0.00)	(0.06)
Budget 1953 × 2 Year	0.03	0.04	−0.00	0.04
	(0.06)	(0.03)	(0.00)	(0.06)
Monsoon 1953 × 2 Year	−0.02	−0.03	0.00	−0.03
	(0.05)	(0.04)	(0.00)	(0.07)
Winter 1953 × 2 Year	0.02	−0.08	−0.00	−0.09
	(0.07)	(0.08)	(0.00)	(0.16)
Pre-Budget 1954 × 2 Year	-0.12^{**}	−0.00	-0.01^{*}	−0.24
	(0.05)	(0.11)	(0.00)	(0.15)
Winter 1952 × 4 Year	−0.01	0.02	−0.00	0.01
	(0.05)	(0.03)	(0.00)	(0.06)
Pre-Budget 1953 × 4 Year	0.00	0.04	0.00	0.05
	(0.04)	(0.04)	(0.00)	(0.05)
Budget 1953 × 4 Year	−0.02	0.03	−0.00	0.00
	(0.06)	(0.03)	(0.00)	(0.06)
Monsoon 1953 × 4 Year	−0.05	−0.01	−0.00	−0.04
	(0.05)	(0.04)	(0.00)	(0.07)
Winter 1953 × 4 Year	−0.04	−0.06	−0.00	−0.17
	(0.06)	(0.08)	(0.00)	(0.15)
Pre-Budget 1954 × 4 Year	−0.06	−0.04	−0.00	−0.19
	(0.04)	(0.06)	(0.00)	(0.12)
Baseline Question Hour	0.06^{***}			
	(0.00)			
Baseline Written Questions		0.09^{***}		
		(0.03)		
Baseline Discussions			0.02^{***}	
			(0.01)	
Baseline Index				0.56^{***}
				(0.07)
Baseline Values	Yes	Yes	Yes	Yes
Adj. R^2	0.56	0.30	0.12	0.42
Num. obs.	1284	1284	1284	1284

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Figure A1: Session-Wise Performance for 4-Year and 6-Year Group



C.3 Effect on Tenures and Career

Column (1) of Table A9 counts the total numbers of terms served by each member. Column (2) indicates the total number of years served by the member after their first term and before 1995. We see that a shorter term does not affect the career trajectory of the members in either groups.

Table A8: Term Lengths and Parliamentary Performance by Session

	Question Hour	Written Questions	Discussions	Index
Budget 1955	0.05	0.03	−0.00	−0.11
	(0.05)	(0.06)	(0.00)	(0.15)
Budget 1956	0.03	0.02	-0.00^{*}	−0.17
	(0.06)	(0.05)	(0.00)	(0.14)
Monsoon 1954	0.10^{*}	0.03	-0.00^{*}	−0.12
	(0.06)	(0.05)	(0.00)	(0.14)
Monsoon 1955	0.11	0.01	-0.00^{*}	−0.12
	(0.07)	(0.05)	(0.00)	(0.14)
Winter 1954	0.06	0.04	-0.00^{*}	−0.13
	(0.06)	(0.06)	(0.00)	(0.14)
Winter 1955	0.09	0.03	−0.00	−0.10
	(0.07)	(0.06)	(0.00)	(0.15)
Budget 1954 × 4 Year	−0.02	0.01	−0.00	−0.01
	(0.04)	(0.04)	(0.00)	(0.15)
Budget 1955 × 4 Year	−0.03	0.02	−0.00	−0.01
	(0.05)	(0.06)	(0.00)	(0.08)
Budget 1956 × 4 Year	−0.08	0.07	−0.00	0.01
	(0.05)	(0.08)	(0.00)	(0.07)
Monsoon 1954 × 4 Year	−0.05	−0.01	−0.00	−0.02
	(0.06)	(0.05)	(0.00)	(0.06)
Monsoon 1955 × 4 Year	−0.06	0.02	−0.00	−0.00
	(0.07)	(0.05)	(0.00)	(0.07)
Winter 1954 × 4 Year	−0.07	−0.00	0.00	0.00
	(0.06)	(0.06)	(0.00)	(0.08)
Winter 1955 × 4 Year	−0.03	0.04	−0.00	0.03
	(0.08)	(0.08)	(0.00)	(0.10)
Baseline Question Hour	0.06^{***}			
	(0.01)			
Baseline Written Questions		0.16^{***}		
		(0.02)		
Baseline Discussions			−0.00	
			(0.00)	
Baseline Index				0.53^{***}
				(0.07)
Baseline Values	Yes	Yes	Yes	Yes
Adj. R^2	0.49	0.52	0.02	0.39
Num. obs.	1001	1001	1001	1001

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Table A9: Term Lengths and Career

	Terms Served	Years Served
	(1)	(2)
2 Year Term	0.11	0.68
	(0.14)	(0.75)
4 Year Term	0.05	0.26
	(0.13)	(0.70)
Control Means	1.69	3.41
State Fixed Effects	Yes	Yes
Observations	214	214
R^2	0.00	0.00

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of shorter terms on careers of members. "2 Years" and "4 Years" indicates a two-year and four-year term respectively.

D Additional Results: Selection on Observables

D.1 Multiple estimation methods

In this section we report the ATE of our combined outcome index using different estimation methods. This includes two regression approaches: the Multi-Regression Imputation (MRI) Estimator that uses two separate regressions, and the Uni-Regression Imputation (URI) Estimator that uses a single regression. Further, we show results from a propensity score weighting estimator and using 1:1 genetic matching.

Table A10 shows that our results remain robust across the estimators, remaining statistically significant at the 1%-5% level for URI, propensity score weighting and genetic matching. Our preferred estimator for the main results in the paper is the Regression (URI), and Table A11 shows that in our chosen method, influential observations do not account for more than 26% of the effect size, exerting no more influence on the results as compared to other methods, with a very small amount of extrapolation.

Table A10: Estimates and Effective Sample Size

Estimator	Estimate	SE	Eff. n	Eff. n Ratio
Regression (MRI)	−1.22	0.79	5.67	0.47
Regression (URI)	−0.69	0.28	10.46	0.87
Propensity Score Weighting	−0.54	0.29	9.12	0.75
Genetic Matching	−0.63	0.12	7.93	0.66

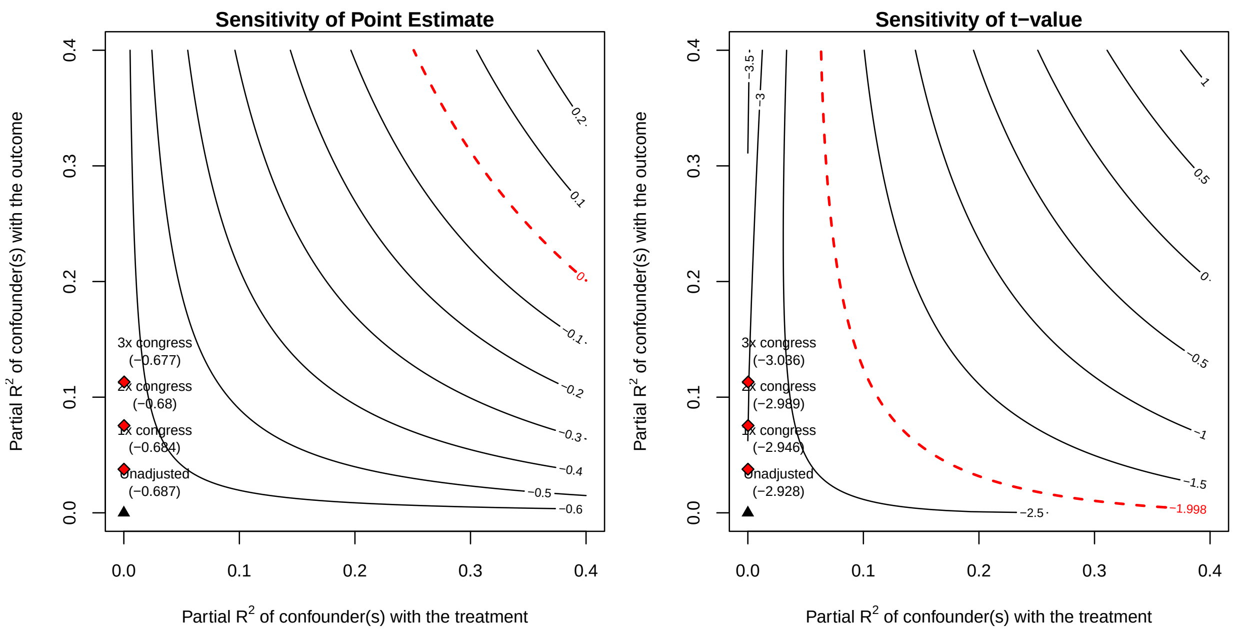
Table A11: Influence and Extrapolation

Estimator	DFBETA Minimum		DFBETA Maximum		Extrapolation	
	Value	Politician	Value	Politician	Control Units	Treatment Units
Regression (MRI)	−0.37	Shahabuddin	0.21	Ganesan	0.15	0.00
Regression (URI)	−0.18	Shahabuddin	0.10	Ganesan	0.00	0.01
Propensity Score Weighting	−0.12	Basu	0.12	Ganesan	NA	NA
Genetic Matching	−0.14	Basu	0.08	Kaul	NA	NA

D.2 Sensitivity Analysis

Figure A2 displays the sensitivity contour plots, showing that the results do not change in substantive interpretation even in the presence of a hypothetical confounder that is three times as influential as membership to the Congress party.

Figure A2: Sensitivity Contour Plots for Outcome Index



D.3 Replacement MPs: Additional Analysis

We produce several additional results for our analysis focusing on the replacement MPs. The main analysis averages over an MP's performance in different stages of their term: an MP serving a 1-year term is only observed during the last year of their tenure, while an MP serving a 5-year term is observed over 5 years, where they can be many years away from the end of their term. In Table A12, we measure outcomes in the last two years of office. Naturally, this analysis can only be conducted for a subset of MPs who served a minimum of two years in office. Second, the analysis may be conflating term lengths with institutional capacity. An MP serving a 5-year term may have more time to gain institutional knowledge and get accustomed to parliamentary procedure. We show that our result remains robust if our treatment group of MPs is only composed of experienced MP: in Table A13, we produce results over the entire period for this subset, while Table A14 shows the results only for the last two years for this subset of MPs.

Table A12: Shorter terms worsen performance of Replacement MPs in the last two years

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
Short Term	−0.08*	−0.46**	−0.03*	−0.58*
	(0.04)	(0.21)	(0.02)	(0.28)
Minister	−0.01	−0.67	0.01	−0.42
	(0.06)	(0.46)	(0.03)	(0.67)
Prior Experience	−0.00	−0.01	0.00	−0.01
	(0.00)	(0.01)	(0.00)	(0.02)
Male	−0.04	0.16	0.01	−0.03
	(0.04)	(0.25)	(0.02)	(0.33)
Post-Graduate	−0.00	0.32	0.02	0.36
	(0.03)	(0.28)	(0.02)	(0.36)
Graduate	0.02	−0.18	0.02	−0.01
	(0.05)	(0.27)	(0.02)	(0.40)
Congress Party	−0.09*	−0.60	−0.03	−0.89*
	(0.05)	(0.35)	(0.02)	(0.50)
Nominated Member	−0.05	−0.99*	0.04	−0.61
	(0.08)	(0.50)	(0.03)	(0.56)
Law Background	−0.01	0.29	−0.00	0.07
	(0.07)	(0.43)	(0.02)	(0.45)
Control Means	0.1	0.73	0.05	0.46
Start-Decade Fixed Effects	Yes	Yes	Yes	Yes
Observations	45	45	45	45
R^2	0.33	0.50	0.49	0.43

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of length of term on measures of legislative performance for MPs elected post death of a sitting member.

Table A13: Shorter terms worsen performance of Replacement MPs with previous experience

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
Short Term	−0.06	-0.47^{**}	−0.01	-0.67^{*}
	(0.04)	(0.18)	(0.02)	(0.34)
Minister	−0.08	-1.21^{**}	−0.04	−1.73
	(0.08)	(0.48)	(0.04)	(1.02)
Prior Experience	−0.00	0.01	0.00	0.02
	(0.00)	(0.02)	(0.00)	(0.03)
Male	0.01	0.41	0.02	0.57
	(0.06)	(0.31)	(0.03)	(0.59)
Post-Graduate	0.02	0.71^{**}	0.03	0.99
	(0.05)	(0.28)	(0.03)	(0.61)
Graduate	0.01	−0.04	0.01	0.08
	(0.05)	(0.22)	(0.02)	(0.46)
Congress Party	−0.01	−0.33	−0.00	−0.41
	(0.07)	(0.31)	(0.03)	(0.65)
Nominated Member	−0.09	-1.38^{**}	0.05	−1.08
	(0.07)	(0.49)	(0.03)	(0.76)
Law Background	0.01	0.17	0.02	0.11
	(0.06)	(0.41)	(0.03)	(0.61)
Control Means	0.11	0.69	0.04	0.59
Start-Decade Fixed Effects	Yes	Yes	Yes	Yes
Observations	36	36	36	36
R^2	0.19	0.60	0.38	0.43

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of length of term on measures of legislative performance for MPs elected post death of a sitting member.

Table A14: Shorter terms worsen performance of Replacement MPs with previous experience in the last two years

	Question Hour	Written Questions	Discussions	Index
	(1)	(2)	(3)	(4)
Short Term	−0.06	−0.57**	−0.04**	−0.68**
	(0.04)	(0.23)	(0.01)	(0.25)
Minister	−0.05	−1.34*	−0.00	−1.20
	(0.09)	(0.65)	(0.03)	(0.73)
Prior Experience	−0.00	0.01	0.00	0.00
	(0.00)	(0.02)	(0.00)	(0.02)
Male	−0.03	0.45	0.01	0.22
	(0.07)	(0.45)	(0.02)	(0.53)
Post-Graduate	0.04	0.76**	0.04	0.93*
	(0.04)	(0.34)	(0.02)	(0.43)
Graduate	0.08	0.19	0.06*	0.59
	(0.07)	(0.38)	(0.03)	(0.49)
Congress Party	−0.10	−0.66	−0.05	−0.99
	(0.07)	(0.43)	(0.03)	(0.64)
Nominated Member	−0.17*	−1.91**	0.08**	−1.30
	(0.09)	(0.64)	(0.03)	(0.80)
Law Background	0.02	0.46	−0.02	0.07
	(0.11)	(0.71)	(0.03)	(0.75)
Control Means	0.1	0.73	0.05	0.46
Start-Decade Fixed Effects	Yes	Yes	Yes	Yes
Observations	27	27	27	27
R^2	0.52	0.71	0.78	0.66

Note: *p<0.1; **p<0.05; ***p<0.01. Estimated effects of length of term on measures of legislative performance for MPs elected post death of a sitting member.